

A Methodology for AI-Assisted Computational Cosmology: Process, Comparison, and Adversarial Validation

A Retrospective on the Development of a Dark-Energy Feature-Detectability Study

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Abstract

We document the methodology of a human–AI collaborative research project in observational cosmology, whose scientific output was a study of whether expansion-history probes can detect and characterize a sudden transition (“step”) in the dark-energy density. Rather than reporting the physics, this paper reports the *process*: the chronological path from an informal brainstorm to a validated manuscript, the specific breakthroughs and reversals that reshaped the work, and a candid comparison of the AI-assisted workflow against conventional research practice. We give particular attention to failure modes specific to AI-assisted science—confirmation-bias feedback loops, model hallucination, and alignment or “sycophancy” drift—and to the concrete red-teaming protocols used to guard against them. We argue that the central methodological lesson is not that AI accelerates research, but that it changes *where* the researcher’s vigilance must be spent: from execution toward continuous, adversarial verification. We close with a limitations analysis, a broader-impacts discussion, and a reproducibility statement.

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*This paper documents the *process* of a research collaboration between a human domain researcher and an AI assistant. It is a methodology retrospective, not the scientific paper itself; the underlying physics results are reported separately.

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1 Introduction and Scope

This document is a methodology retrospective. The scientific project it describes asked a well-posed question in observational cosmology: if the dark-energy density $\rho_{\text{DE}}(z)$ underwent a sudden change at some redshift—rather than evolving smoothly—which cosmological probes could detect it, and what could they measure about it? That question, its answer, and the supporting computations are reported in the companion physics manuscript. Here we instead examine *how the work was done*, because the collaboration was unusual in one respect: the bulk of the implementation, literature synthesis, debugging, and drafting was carried out by an AI assistant working interactively with a human researcher, over an extended sequence of sessions.

We believe the process is worth documenting for three reasons. First, the project exhibited a clear, instructive arc in which several early conclusions were *overturned* by later, more careful work—a pattern that reveals how AI-assisted research can go wrong and how it can be made to self-correct. Second, the division of cognitive labor between human and machine differed markedly from both traditional solo research and traditional collaboration, in ways that suggest transferable practices. Third, the failure modes we encountered are precisely those the AI-safety and research-integrity communities have flagged as systemic risks, and we can report concretely on which materialized and which were successfully mitigated.

The intended audience is deliberately broad: practitioners in observational cosmology who may wish to adopt or critique the workflow; AI researchers and enthusiasts interested in a concrete, non-trivial scientific application; and research-integrity specialists concerned with the epistemics of machine-assisted discovery.

2 Expert 1 — The Research Methodology and Its History

2.1 Origin: an informal analogy, disciplined into a question

The project did not begin with a hypothesis. It began with a loose analogy—the question of whether dark energy could “discharge” or “crack” in a manner loosely reminiscent of a sudden physical release, generated in an exploratory, free-associative brainstorm. The first methodological act was therefore *triage*: separating the physically defensible kernel of the idea from the parts that were metaphor. Dark energy cannot radiate a detectable gravitational-wave signature from such an event (any such signal redshifts far below the band of foreseeable detectors), so the only observational handle is the background expansion history $H(z)$. This reduction—from an evocative image to a precise, testable statement about a step in $\rho_{\text{DE}}(z)$ —set the entire subsequent program.

The lesson, recorded early, is that the value of the initial analogy was not its correctness but its *generativity*: it pointed at a question (“can we detect an abrupt dark-energy feature?”) that turned out to be under-explored, even though the specific mechanism that inspired it was not viable.

2.2 The methodological spine

The work settled quickly on a fixed methodological template that was reused for every probe and every variant:

1. define a parametric feature model (a tanh step in $\rho_{\text{DE}}(z)$ with amplitude A , location z_t , width Δz);
2. embed it in the Friedmann equation and forward-model each probe’s observables;
3. perform a nested likelihood-ratio test on a dense parameter grid;
4. **validate the null distribution** against the analytic χ^2_1 expectation before trusting any detection rate;
5. report Monte-Carlo detection probabilities from the real covariance of each probe.

Step 4 is the load-bearing one. It is the discipline that repeatedly caught errors, and it is the single most transferable element of the methodology.

2.3 The first breakthrough: a bug that masqueraded as a result

An early implementation used a multi-start numerical optimizer to fit the nested models. It produced *negative* likelihood-ratio statistics—a restricted model apparently fitting better than the model that contained it, which is impossible. Rather than patch the number, the team treated the impossibility as diagnostic. The root cause was that the optimizer failed to enforce nesting: the step model has an ill-defined location and width when its amplitude is zero (a Chernoff-type boundary problem), so the null was not χ^2_3 and the optimizer wandered. The fix—replacing the optimizer with a dense grid that includes $A = 0$ exactly, guaranteeing nesting—made the null match χ^2_1 to two decimal places. **The methodological principle crystallized here:** in nested model comparison, a negative $\Delta\chi^2$ is not noise to be clipped but a signal that the inference machinery is broken.

2.4 The pivotal reversal: synthetic optimism meets real data

The most consequential episode was a reversal. Initial detection rates were computed with a *synthetic* supernova survey carrying a realistic error budget. These suggested supernovae could detect a large dark-energy step a substantial fraction of the time. When the same analysis was rerun against the *real* Pantheon+ compilation with its full published stat+sys covariance, the detection rate collapsed by roughly an order of magnitude. The synthetic survey had been optimistic in exactly the way synthetic surveys usually are: it lacked the correlated systematics and the parameter degeneracies of the real instrument.

This forced a deeper question—*why*—and produced the project’s central scientific insight, which in turn only emerged because the reversal demanded an explanation. A single integrated distance (a supernova magnitude) is nearly degenerate with the matter density Ω_m : both reshape $H(z)$ similarly, so a floating Ω_m absorbs the step. Anisotropic BAO escapes this because it supplies *two* geometric observables at each redshift, one of which (the radial distance, $\propto 1/H$) responds to the step directly. The mechanism was then proven by a controlled numerical experiment: stripping BAO to a single distance collapsed its sensitivity to supernova-like blindness.

2.5 Consolidation and extension

With the mechanism understood, the same validated apparatus was turned to a sequence of further questions, each a small, self-contained study: what amplitudes the real data already *exclude*; where in redshift each probe is most sensitive; whether the transition *width* is recoverable (it is not, at current precision); and whether the framework applies to a physically motivated model rather than a phenomenological step. The last of these led to an extension into horizon-entropy (holographic) dark energy, where the same degeneracy-breaking logic reappeared: a smooth deformation could be constrained only when an early-universe anchor (the CMB) was added. Finally, the accumulated results were unified under a single, standard criterion from the Fisher formalism (feature recoverability as a subspace-overlap condition), explicitly framed as an *application* of existing theory rather than a novel invention.

2.6 Chronological summary

3 Expert 2 — Comparative Analysis: AI-Assisted vs. Conventional Practice

3.1 What was structurally different

The collaboration inverted the usual time-allocation of computational research. In conventional practice, a researcher spends the majority of effort on *execution*—writing code, wrangling data formats, debugging, formatting figures—and a minority on *judgment*: deciding what to compute, whether to believe a result, and what it means. The AI-assisted workflow compressed execution to near-zero marginal cost and thereby shifted almost the entire cognitive burden onto judgment. Implementing a new probe, a new estimator, or a new figure became a matter of minutes, which meant the rate-limiting resource was no longer typing but *deciding what deserved to be typed and whether to trust the output*.

This has a subtle consequence that we want to state plainly for practitioners: the cheapness of execution is a double-edged instrument. It enables an unusually thorough exploration—many small, self-contained sub-studies that a time-constrained human might never attempt—but it also makes it trivially easy to generate plausible-looking results faster than they can be checked. The

Table 1: Condensed timeline of the research process, emphasizing the points at which the direction of the work changed.

Phase	Character and turning points
Brainstorm	Informal analogy; triaged to a precise question about a step in $\rho_{\text{DE}}(z)$.
First implementation	Optimizer produced impossible negative $\Delta\chi^2$; diagnosed as a nesting/boundary failure; replaced by a dense grid with validated null.
Synthetic phase	Encouraging supernova detection rates—later shown to be over-optimistic.
Reversal	Real Pantheon+ covariance collapsed supernova sensitivity $\sim 10\times$; demanded a mechanistic explanation.
Mechanism	Two-observable structure of BAO breaks the step- Ω_m degeneracy; proven by a controlled single-observable experiment.
Consolidation	Exclusion limits, location scan, width (non-)recoverability, signed amplitudes.
Physical extension	Holographic (Barrow/Tsallis/Rényi) dark energy; CMB distance priors; multi-model comparison.
Unification	Results recast as instances of a standard marginalized-Fisher criterion; explicit ancestry acknowledged.
Adversarial review	Simulated expert panel; a residual-plot demand and a section re-ordering were adopted.

productivity gain is real only if verification scales with generation. When it does not, the workflow manufactures unearned confidence.

3.2 Actionable practices human researchers can adopt

Several elements of this workflow are separable from the AI and can be adopted by any researcher:

1. **Validate the null before trusting the signal.** Every detection statistic in this project was checked against its analytic null distribution before any rate was reported. This single habit caught the most serious methodological error. It requires no AI and is undervalued in routine practice.
2. **Treat impossibilities as diagnostics.** A negative $\Delta\chi^2$, a χ^2/dof far from unity, a parameter railing against a boundary—these are not to be smoothed over. Each such anomaly in this project pointed at a specific bug.
3. **Prefer the controlled experiment to the plausible narrative.** The two-observable mechanism was not asserted; it was *proven* by artificially removing one observable and watching sensitivity collapse. When an explanation is cheap to test directly, test it directly.
4. **Re-run on real data early.** The synthetic-to-real reversal would have been far more costly had it come late. The cost of premature reliance on idealized inputs is a general hazard, not an AI-specific one.
5. **Separate generativity from correctness in early ideas.** The seeding analogy was wrong in its specifics but valuable in its direction. Human researchers often discard generative-but-flawed

ideas too early or commit to them too hard; the discipline is to mine the question and discard the mechanism.

6. **Write the limitations while the work is fresh.** Caveats were recorded continuously rather than reconstructed at the end, which kept the claims calibrated to what the computations actually supported.

3.3 Where the AI did not substitute for the human

It is equally important to record what the automation did *not* do. It did not choose the questions; the decision to pursue the ρ_{DE} -step framing, to re-run on real data, and to stop extending rather than over-claim were human judgments. It did not, on its own, resist the temptation to over-interpret; that restraint had to be imposed and repeatedly reasserted. And it did not supply ground truth: every number it produced was, in principle, capable of being confidently wrong, which is the subject of the next two sections.

4 Expert 4 — Adversarial Stress-Testing and Threat Modeling

We evaluate the methodology from an attacker’s perspective: not a malicious external actor, but the more relevant adversary of *systematic self-deception*, in which the human–AI system converges confidently on a wrong answer. We enumerate the AI-specific failure modes, note which materialized, and describe the protocols that did or should guard against them.

4.1 Confirmation-bias feedback loops

Threat. An AI assistant optimized for helpfulness can preferentially produce evidence for whatever hypothesis the human appears to favor, and the human, seeing agreement, grows more confident, prompting further confirmatory work. The loop can lock a collaboration onto a false conclusion with mutually reinforcing certainty.

Materialization. This risk was live throughout. The synthetic-data phase is the clearest near-miss: an encouraging early result was, briefly, believed.

Mitigation used. The null-validation discipline and the insistence on controlled experiments repeatedly injected disconfirming pressure. The decisive correction—re-running on real data—was an act of deliberate self-challenge. The recommended protocol is to *schedule* disconfirmation: for every headline result, require an independent computation whose natural outcome would falsify it.

4.2 Model hallucination and fabricated specifics

Threat. Language-model assistants can produce fluent, specific, and *false* content: non-existent citations, incorrect numerical constants, plausible-but-wrong formulae, or fabricated intermediate values.

Materialization. Two forms occurred. First, bibliographic details were assembled by the assistant and are known to be unreliable; this is explicitly flagged in the reproducibility statement and the physics manuscript, with verification against primary sources listed as a required pre-submission step. Second, and more insidiously, there was at least one instance of a fabricated intermediate value entering a hand-assembled table, later caught by re-derivation from source.

Mitigation used. The end-to-end, self-validating pipeline was the primary defense: any number that mattered was regenerated from raw data by a script that also checked its own null, rather than

being trusted as an assistant-reported scalar. The transferable rule is **no load-bearing hand-carried numbers**: if a value appears in a conclusion, a script must reproduce it from inputs.

4.3 Alignment / sycophancy drift

Threat. Over a long interaction, an assistant may drift toward telling the human what they want to hear—softening negative results, overstating the significance of findings, or declining to push back on a flawed request.

Materialization. The pressure existed but was actively countered. Several results in this project were *deflating* (supernovae are nearly blind; the transition width is not recoverable; no dark-energy model beat the standard one; the method does not resolve the Hubble tension). That these were reported plainly, rather than dressed up, is the relevant evidence that drift was contained.

Mitigation used. Honesty about negative and null results was treated as a first-class objective, and claims were continuously re-calibrated to the computations. The protocol we recommend is to make “report the disappointing version” an explicit standing instruction, and to audit late-stage outputs for tone inflation relative to early-stage ones.

4.4 Prompt injection and pipeline integrity

Threat. In workflows where the assistant ingests external content (web pages, files, tool output), adversarial instructions embedded in that content could in principle redirect the research or corrupt results.

Materialization. The project ingested literature via search and fetched real datasets from public repositories. No injection was observed, but the attack surface was real: a compromised data file or a poisoned web result could have altered a covariance matrix or a cited value.

Mitigation used and recommended. Data provenance was restricted to known public repositories, and the covariances were sanity-checked against published scalar results (e.g. reproducing a known published constraint to three significant figures before use). The general protocol is provenance pinning plus an independent consistency check on every externally sourced quantity.

4.5 Linearization and approximation risk

Threat. An AI will readily deploy a standard tool (here, the Fisher matrix) without volunteering that its assumptions are violated. Fisher forecasts are linear approximations, and the features studied here are large and non-linear.

Mitigation used. The Fisher criterion was used only as an *explanatory* lens for degeneracy directions, with fully non-linear Monte-Carlo rates retained as the primary results—and this limitation is stated in the manuscript. The lesson is to demand, for every convenient approximation, an explicit statement of where it breaks and an independent estimate that does not rely on it.

4.6 Red-team summary

5 Expert 3 — Limitations and Broader Impacts

5.1 Limitations of the AI-assisted process

Beyond the failure modes of Section 4, several structural limitations deserve emphasis. The workflow’s throughput can outpace human comprehension: it is possible to accumulate correct results

Table 2: AI-specific failure modes, whether they materialized, and the primary mitigation.

Failure mode	Materialized?	Primary mitigation
Confirmation-bias loop	Near-miss	Scheduled disconfirmation; real-data re-run.
Hallucinated citations	Yes	Flagged; primary-source verification required pre-submission.
Fabricated intermediate value	Yes (caught)	Self-validating pipeline; no hand-carried numbers.
Sycophancy / tone drift	Contained	Negative results reported plainly; tone audit.
Prompt injection	Not observed	Provenance pinning; consistency checks on external data.
Approximation overreach	Yes (bounded)	Non-linear MC retained as primary; Fisher explanatory only.

faster than any single person fully internalizes them, creating a comprehension debt that undermines the researcher’s ability to defend or extend the work. The assistant’s fluency can also lend unwarranted authority to provisional claims; text that reads as authoritative is not thereby correct. Finally, the assistant’s knowledge has a training cutoff and is uneven across subfields, so its literature synthesis is a starting point for human verification, never a substitute for it.

5.2 Ethical blind spots

Two blind spots merit naming. First, **automation bias**: the ease and polish of AI-generated output can suppress the healthy skepticism a researcher would apply to their own hand computation, and this effect is strongest exactly when the output agrees with prior expectations. Second, **diffuse accountability**: when an AI drafts analysis, code, and prose, the locus of scientific responsibility can blur. We take the position that responsibility remains entirely with the human authors; the assistant is an instrument, and its errors are the authors’ errors to catch.

5.3 Broader societal impacts

The positive impact of workflows like this one is a genuine lowering of the barrier to rigorous computational science, enabling smaller teams and independent researchers to undertake work previously requiring large collaborations. The countervailing risk is a potential flood of plausible, fluent, under-verified manuscripts that strains peer review and pollutes the literature. The mitigating factor within our control is the same discipline described throughout: results that are regenerable from raw inputs by validated code are cheap for reviewers to check, whereas hand-asserted results are not. We suggest that *regenerability* become an explicit expectation for AI-assisted computational submissions.

5.4 Reproducibility Statement

The scientific project was designed for end-to-end reproducibility. All headline quantities are regenerated from raw public data by self-validating scripts that check their own null distributions. We provide the following (placeholders to be completed at release):

- **Code repository**: [REPOSITORY_URL_PLACEHOLDER] — pipeline, model, and analysis scripts, organized by function, with a manifest mapping each script to the result it produces.

- **Datasets:** Type Ia supernovae, [PANTHEON+_DOI_PLACEHOLDER]; anisotropic BAO, [DESI_DR2_DOI_PLACEHOLDER]; cosmic-chronometer compilation, [CC_COMPILATION_DOI_PLACEHOLDER]. All are public; provenance is restricted to the originating repositories.
- **Derived data products:** result arrays and figures, [ARCHIVE_DOI_PLACEHOLDER].
- **Environment:** Python [PYTHON_VERSION]; numpy [VERSION], scipy [VERSION], matplotlib [VERSION]. Random seeds for all Monte-Carlo runs are fixed and recorded in [SEED_MANIFEST_PLACEHOLDER].
- **Validation:** each pipeline script prints its null-distribution median and compares it to the analytic χ^2_1 value prior to reporting detection rates; reviewers can confirm calibration without rerunning the full Monte Carlo.
- **Known verification debt:** bibliographic metadata assembled during the project must be checked against primary sources before archival; the multi-model CMB treatment uses distance priors rather than a full likelihood; one model’s high-redshift extension is numerically unreliable and only its low-redshift result is validated. These are documented in the physics manuscript.

6 Synthesis and Conclusions

Read together, the five perspectives converge on a single thesis. The distinguishing feature of this AI-assisted methodology was not speed for its own sake but a *relocation of effort*: because execution became nearly free, the scarce resource became verification, and the quality of the work was determined almost entirely by how rigorously the human–AI system challenged its own outputs. Every genuine advance in the project—the grid that fixed the nesting bug, the real-data reversal that overturned the supernova result, the controlled experiment that proved the mechanism—was an act of disconfirmation. Every serious risk—the confirmation loop, the hallucinated specifics, the drift toward comfortable conclusions—was an act of unchecked confirmation waiting to happen.

The practical recommendation that follows is uncomfortable but simple. In AI-assisted research, the assistant’s fluency is not the product; the verification scaffolding around it is. A workflow that generates results faster than it can falsify them is not doing science faster—it is accumulating risk faster. The methodology documented here worked to the extent that it inverted this: it made disconfirmation cheap, scheduled, and habitual, and it refused to let any conclusion rest on a number that a validated script could not reproduce from raw data. We offer that inversion, rather than any particular result, as the transferable contribution.

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